

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Messenger RNA (mRNA) and transfer RNA (tRNA) are involved in protein synthesis.

(a) The diagram shows the sequence of bases in part of a DNA template (antisense) strand and the mRNA molecule formed during protein synthesis.

DNA template strand

A	T	G	C	C	G	T	T
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mRNA

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(i) Complete the diagram to show the sequence of bases in this mRNA.

(1)

(ii) Name this stage of protein synthesis.

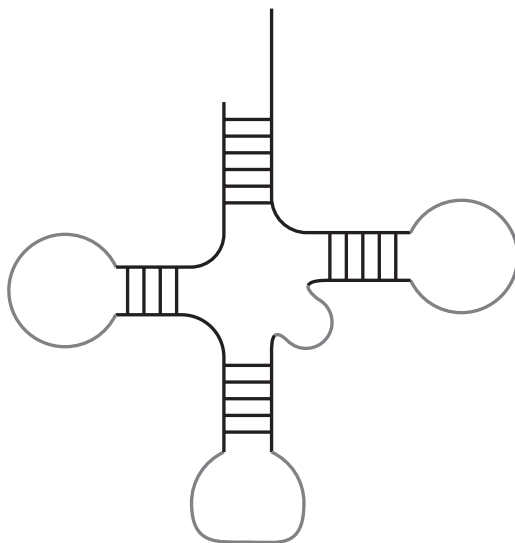
(1)

(iii) Name the enzyme used in this stage of protein synthesis.

(1)



(b) The diagram shows part of a tRNA molecule.



Compare and contrast the structure of an mRNA molecule with that of a tRNA molecule.

(3)

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(Total for Question 1 = 6 marks)